

Panduan Setup Google Analytics 4 (GA4)

Rizal Heritage — dari nol sampai bisa digunakan

There are **two separate pieces** here, and both need to be done for GA4 to actually be useful:

- 1 **Tracking** — a small script (`gtag.js`) added to every page that sends real visitor data to Google. Without this, your GA4 property just sits empty forever, no matter what else you configure.
- 2 **Reporting** — a read-only service account this app uses to pull GA4's stats back into `/admin/insight`, so you don't have to leave the admin panel to see them.

Part 1 alone gives you the full Google Analytics website (`analytics.google.com`) with all its normal reports.

Part 2 is optional — only do it if you want a summary of the same numbers surfaced inside `/admin/insight` too.

Part 1 — Create the GA4 property and get tracking working

1. Create a Google Analytics account + property

- 1 Go to `analytics.google.com` and sign in with any Google account (doesn't need to be tied to Hosting/GitHub/etc — a personal or business Gmail is fine).
- 2 **Admin** (gear icon, bottom-left) → **Create Account** if you don't have one yet. Name it something like "Rizal Heritage".
- 3 Inside that account, **Create Property** → name it "Rizal Heritage Website" → set your timezone (Indonesia) and currency (IDR) → **Next**.
- 4 Fill in the small business-details questionnaire (industry category, business size — doesn't affect anything technical, just Google's own categorization) → **Create**. Accept the Terms of Service.

2. Create a Web data stream and get the Measurement ID

- 1 Still in **Admin**, under the property you just made → **Data Streams** → **Add stream** → **Web**.
- 2 **Website URL**: `https://rizalheritage.com`. **Stream name**: anything, e.g. "Rizal Heritage — main site".
- 3 Click **Create stream**. You'll land on the stream's details page — at the top right is your **Measurement ID**, formatted like `G-XXXXXXXXXX`. Copy it.

3. Add the Measurement ID to the app

The app already has the tracking snippet built in (`resources/views/app.blade.php`) — it just needs the ID to activate. On your **server** (`~/laravel-app/.env`):

```
nano ~/laravel-app/.env
```

Add (or fill in if the line's already there from the template):

```
GA4_MEASUREMENT_ID=G-XXXXXXXXXX
```

Then:

```
cd ~/laravel-app
php artisan config:clear
```

That's it — no rebuild needed, this isn't part of the Vite frontend bundle (it's rendered server-side in the base HTML template), so no `npm run build/upload` step required for this specific change.

4. Verify it's actually sending data

- 1 Open `https://rizalheritage.com` in a normal browser tab (not the `/etc/hosts` trick from earlier — use the real domain now that DNS/SSL are live).
- 2 In GA4, go to **Reports** → **Realtime**. Within ~30 seconds you should see **1 active user** — that's you. Click around the site a bit and watch the events (`page_view`, etc.) show up in Realtime.
- 3 If nothing shows up: open your browser's DevTools → Network tab, reload the page, and search for a request to `google-analytics.com/g/collect`. If it's missing entirely, double check `GA4_MEASUREMENT_ID` is set and `config:clear` was run; if it's present but GA4 doesn't show it, an ad blocker/privacy extension in your test browser is likely eating it — try an incognito window with extensions disabled, or a phone on mobile data.

That's the whole tracking setup. Full reports (Acquisition, Engagement, Demographics, etc.) take 24–48 hours to populate with meaningful data — Realtime is immediate, the rest of GA4 batches overnight.

Part 2 (optional) — Pull GA4 stats into `/admin/insight`

This app's own `/admin/insight` dashboard already works without this — it's backed by this app's own visit/search tracking (a separate system, stored in your MySQL database). This part just adds a GA4-sourced section on top, showing the same kind of numbers (active users, sessions, top countries, device breakdown) but computed by Google instead.

5. Enable the Analytics Data API in Google Cloud

- 1 Go to `console.cloud.google.com`. You can use the same Google account as Part 1, or a different one — GA4 property access is granted separately in step 7 regardless.
- 2 If you don't have a project yet: top bar → **Select a project** → **New Project** → name it "Rizal Heritage Analytics" → **Create**.
- 3 With that project selected, go to **APIs & Services** → **Library**, search "Google Analytics Data API", open it, click **Enable**.

6. Create a service account and download its key

- 1 **APIs & Services** → **Credentials** → **+ Create Credentials** → **Service account**.
- 2 Name it something like `ga4-reporter` → **Create and Continue** → skip the optional role/access steps (not needed — access is granted directly in GA4 instead, next step) → **Done**.
- 3 Click into the service account you just created → **Keys** tab → **Add Key** → **Create new key** → **JSON** → **Create**. A `.json` file downloads to your computer — **keep it, don't lose it, don't commit it to Git**.
- 4 Note the service account's email from that same page — it looks like `ga4-reporter@your-project-id.iam.gserviceaccount.com`.

7. Grant that service account access to your GA4 property

- 1 Back in analytics.google.com → **Admin** → under the property → **Property Access Management**.
- 2 + (top right) → **Add users** → paste the service account's email from step 6.4 → role: **Viewer** (read-only — that's all this app needs) → **Add**.

8. Get the values .env needs

Open the downloaded JSON key file — you need exactly two fields out of it (never paste the whole file anywhere):

```
{
  "client_email": "ga4-reporter@your-project-id.iam.gserviceaccount.com",
  "private_key": "-----BEGIN PRIVATE KEY-----\nMIIEvQIBADANBgk...\n-----END PRIVATE KEY-----\n"
}
```

You also need the **Property ID** (a plain number, *not* the Measurement ID from Part 1): GA4 → **Admin** → under the property → **Property details**, near the top — looks like `properties/123456789`, you only need the number `123456789`.

9. Add these to .env on the server

```
nano ~/laravel-app/.env
```

```
GA4_PROPERTY_ID=123456789
GA4_CLIENT_EMAIL=ga4-reporter@your-project-id.iam.gserviceaccount.com
GA4_PRIVATE_KEY="-----BEGIN PRIVATE KEY-----\nMIIEvQIBADANBgk...\n-----END PRIVATE KEY-----\n"
```

Important: **keep it as one line, wrapped in double quotes, with literal `\n` characters** exactly as they appear in the JSON file (don't turn them into real line breaks) — `Ga4Client.php` un-escapes them back into a real multi-line key internally.

Then:

```
cd ~/laravel-app
php artisan config:clear
```

10. Verify

Log into `/admin` as an admin user, open **Insight**. The GA4 section should now show real numbers instead of "not configured". It may show zeros for the first day if Part 1 was only just switched on — GA4 needs a little real traffic before there's anything to report.

Summary — what each variable actually does

.env variable	What it's for	Required for
<code>GA4_MEASUREMENT_ID</code>	Public tag, activates the <code>gtag.js</code> snippet on every page	Any GA4 tracking at all (Part 1)

.env variable	What it's for	Required for
GA4_PROPERTY_ID	Which GA4 property to read stats from	/admin/insight's GA4 section (Part 2)
GA4_CLIENT_EMAIL	Service account identity	same
GA4_PRIVATE_KEY	Service account's signing key	same

You can do Part 1 without Part 2 (tracking works, just no summary inside the admin panel). Part 2 without Part 1 is pointless — there'd be nothing for it to read.